

Forecasting Climate Change and Diabetes Risk: Time Series and Logistic Regression Analysis

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Abstract—This report presents the findings of two statistical analyses conducted on separate datasets. The first analysis involves time series analysis on two data files provided by the Climate Institute of the University of East Anglia. The second analysis utilizes data collected from an Iraqi University Hospital in 2020 to develop a binary logistic regression model for diabetes diagnostic. The report details the methodology used for each analysis, including preliminary assessments, model estimations, and evaluations. The findings of the analyses are summarized, and the suitability of the chosen models is discussed. Generally, the model ARIMA for both time series were proposed for forecasting the average temperature in the future, while the logistic regression also performed an excellent accuracy with 96.84%.

Index Terms—Time Series Analysis, Logistic Regression, Modeling, Diabetes, Climate forecast

I. INTRODUCTION

Time series analysis and logistic regression are two essential techniques used in data science and statistical modeling for making predictions and understanding complex phenomena. In this report, we will focus on applying these techniques to two different datasets: the Climate Institute of the University of East Anglia's temperature data and a diabetes dataset collected from an Iraqi University Hospital in 2020.

The first part of this report focuses on time series analysis of temperature data. Temperature is a fundamental climate variable [1], and the study of its time series can provide valuable insights into climate change, variability, and trends [1].

The objective of the time series analysis is to identify suitable models to estimate the temperature time series and forecast the temperature for the year 2004 using data up to and including 2003 as the training set. The study will consider different models, such as exponential smoothing, ARIMA/SARIMA, and simple time series models, and use diagnostic tests to select the most suitable model for forecasting purposes [2].

The second part of this report focuses on logistic regression. Millions of individuals worldwide suffer from the chronic disease diabetes, which is a major source of morbidity and mortality [3]. Logistic regression is a type of regression analysis used to predict binary outcomes, and it is often used in medical research, finance, and marketing. In this study, we will use a diabetes dataset collected from an Iraqi

University Hospital in 2020 to predict the likelihood of a patient having diabetes based on blood test results. The dataset contains twelve variables related to blood tests, such as urea, creatinine ratio, average blood glucose levels, cholesterol levels, and body mass index, among others [4]. We will use the exploratory data analysis to understand the variables and transform them, if necessary, to develop a suitable binary logistic regression model [5].

The objective of this part of the study is to estimate a binary logistic regression model to predict diabetes based on blood test results, ignoring the 'prediabetic' cases in the dataset. We will split the dataset into a training and a test set and evaluate the models' performance on the test set using a confusion matrix. We will also test the final model on the 'prediabetic' cases and estimate the probability that these cases will be diagnosed as diabetic.

The use of time series analysis and logistic regression in these two different datasets highlights the versatility and applicability of these techniques in different domains. These techniques can be used to analyze complex phenomena and make predictions, which can be useful in various fields, such as climate science, healthcare, finance, and marketing.

In summary, this report aims to provide a comprehensive analysis of time series analysis and logistic regression techniques applied to two different datasets. The study will focus on identifying suitable models for the temperature time series and developing a binary logistic regression model to predict diabetes based on blood test results. The results of this study will provide valuable insights into the versatility and applicability of these techniques in different domains.

II. TIME SERIES ANALYSIS

A. Data sources and Methodology

1) *Data sources:* We have two datasets. One dataset contains the monthly average temperature records of Armagh from January 1844 to December 2004. The other dataset provides a simplified version of the same data, showing the yearly average temperatures from 1844 to 2004.

For this study, I am using the data from 1984 to 2003 as the training set, while the data for the year 2004 is used as the testing set.

2) Methodology:

• Exponential Smoothing (ES):

The ETS (error, trend, seasonality) framework is utilized to automatically determine the most suitable ES approach for each series based on the data requirements. This method involves smoothing the historical observations of the series, with exponentially more weight given to the most recent observations. The chosen techniques range from straightforward exponential smoothing that only takes into account level variations to more advanced techniques that take into account both seasonality as well as patterns of various types.

• Autoregressive Integrated Moving Average (ARIMA):

This approach aims to determine the optimal elements and parameters of the ARIMA model for each series. The Augmented Dickey-Fuller differencing test is used to determine the (seasonal) differencing parameters, d/D . Subsequently, different ARIMA methods are fitted within ranges of the parameters p/P , the number of (seasonal) lags, and q/Q , the order of (seasonal) moving averages. The Akaike Information Criterion is then used to determine the model that best fits the data.

• Simple time series models (Mean Model, Seasonal Naive Model)

The simple time series models, including the mean model and seasonal naive model, are basic approaches used in time series analysis. The mean model assumes that the future value of the time series will be the same as the historical mean of the series. This model is suitable for time series data that do not exhibit any trend or seasonality. The mean model can be written mathematically as:

$$\hat{y}_{t+1} = \bar{y} \quad (1)$$

where $\hat{y}_{t+1}$ is the predicted value of the time series at time $t + 1$, and $\bar{y}$ is the mean of the historical data.

The seasonal naive model, on the other hand, assumes that the future value of the time series will be the same as the most recent observed value from the same season. This model is useful for time series data that exhibit strong seasonal patterns. The seasonal naive model can be written mathematically as:

$$\hat{y}_{t+1} = y_{t+1-m} \quad (2)$$

where $\hat{y}_{t+1}$ is the predicted value of the time series at time $t + 1$, y_{t+1-m} is the most recent observed value from the same season, and m is the length of the seasonal cycle.

Both the mean model and seasonal naive model are straightforward and easy to implement, but they are often outperformed by more sophisticated models that take into account trends, seasonality, and other factors that affect the behavior of the time series.

3) *Evaluation metrics:* The matrices including MAPE, RMSE, and MASE are taken into consideration to evaluate

the forecasting performance of various decomposition bagging ARIMA, Seasonal ARIMA, ETS models, as well as basic time series models. Furthermore, the highly regarded measurements MAPE and RMSE compute the bias in absolute percent and squared forms.

B. Assessment of the raw time series

1) *Data Analysis:* The first step in ensuring the integrity of our data analysis involves checking for missing or NaN values in the dataset, and this is precisely what we did. The summary of the dataset is presented in Table I. After reviewing the summary, we confirmed that both the minimum and maximum temperatures in both the monthly and yearly datasets were within practical ranges. However, we discovered that the yearly dataset was non-stationary, which led us to transform it using the `diff(Yearly_data, differences = 2)` function.

By conducting this initial data quality check, we have established a solid foundation for further analysis and model building, which can be carried out with greater confidence.

TABLE I
MONTHLY AND ANNUAL AVERAGE TEMPERATURE SUMMARY

Statistics	Monthly	Annual
Count	1932	161
Minimum	0.300	6.700
1st Quartile	5.300	8.200
Median	8.300	8.500
Mean	8.534	8.489
3rd Quartile	12.100	8.800
Maximum	17.200	9.700

2) *Data Visualization:* In this section, we are focusing on visualizing the monthly and yearly time series data.

We have created a plot for the average temperature for both the monthly and yearly data which is shown in Figure 1. This plot helps us to identify the trend and seasonality present in the monthly data.

Regarding the yearly plot, we have calculated the difference between the yearly temperatures observed in consecutive years. We have noticed that the plot is not stationary around the mean value figure 1. This means that there is a trend present in the yearly data that needs to be accounted for while building a suitable time series model.

C. Investigation of suitable models

• Monthly data

– Exponential Smoothing Model

ETS(A,A,A) is a type of exponential smoothing model used for time series forecasting. The model is based on three components of a time series: level, trend, and seasonality.

In this ETS(A,A,A) model, the smoothing parameters (figure 4) are $\alpha = 0.0267$, $\beta = 1e-04$, and $\gamma = 1e-04$. The initial states are also specified, with the level starting at 9.0553, the trend at -0.0013, and the seasonality at various values for each month. The sigma value is 1.2163, which represents the

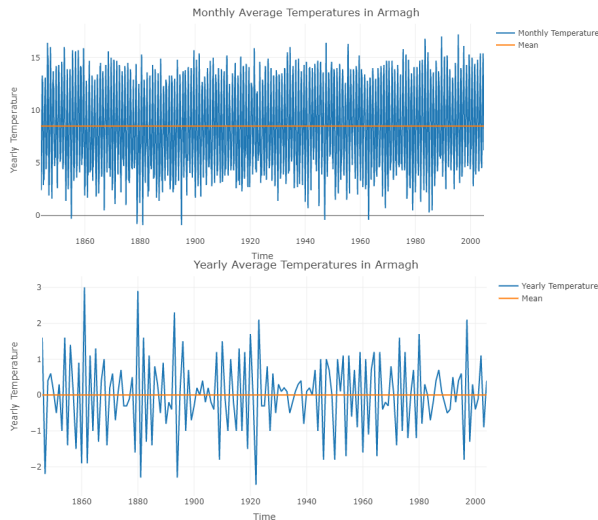


Fig. 1. Plot of monthly and yearly average temperature from 1844 to 2004

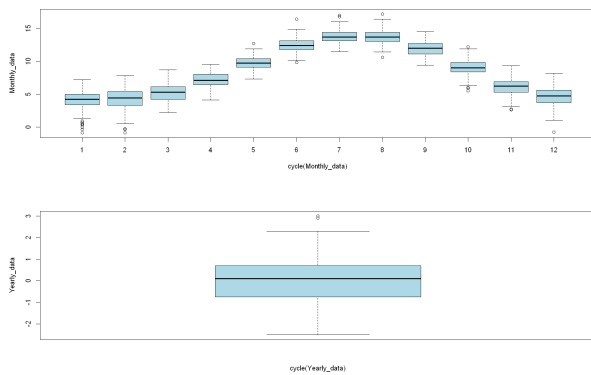


Fig. 2. Boxplot of monthly and yearly average temperature from 1844 to 2004

estimated standard deviation of the residuals (Figure 4).

The training set error measures show that the model has a mean absolute error (MAE) of 0.9465906 and a root mean squared error (RMSE) of 1.211239, indicating that the model is performing reasonably well on the training data. The Ljung-Box test shows that the residuals are not autocorrelated, indicating

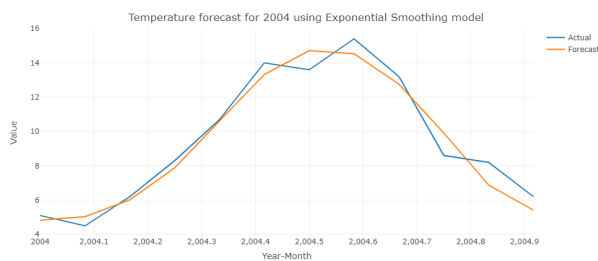


Fig. 3. Monthly Data ETS model Forecast

```
ETS(A,A,A)

Call:
ets(y = Monthly_train_data)

Smothing parameters:
  alpha = 0.0267
  beta  = 1e-04
  gamma = 1e-04

Initial states:
  l = 9.0553
  b = -0.0013
  s = -3.9238 -2.4576 0.5863 3.4298 5.2013 5.384
      4.0052 1.2916 -1.4396 -3.3028 -4.2866 -4.4879

sigma: 1.2163

AIC   AICc   BIC
15285.27 15285.59 15379.79

Training set error measures:
      ME   RMSE   MAE   MPE   MAPE   MASE   ACF1
Training set 0.01770418 1.211239 0.9465906 -Inf   Inf  0.7096361 0.1908923

Ljung-Box test

data: Residuals from ETS(A,A,A)
Q* = 108.35, df = 24, p-value = 1.093e-12

Model df: 0. Total lags used: 24
```

Fig. 4. Monthly Data ETS model Summary

A matrix: 2 × 8 of type dbl

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Thell's U
Training set	0.01770418	1.2112392	0.9465906	-Inf	Inf	0.7096361	0.1908923	NA
Test set	0.17450272	0.7804156	0.6675100	1.792123	7.67026	0.5004161	-0.4164430	0.3343079

Fig. 5. Monthly Data ETS model Accuracy

that the model is a good fit for the data.

Finally, the model generates point forecasts for the next twelve months, along with corresponding 80% and 95% prediction intervals (Figure 5).

– SARIMA Model

The output shows the result of fitting an ARIMA model to monthly train data. The model has non-seasonal order (1,0,2) and seasonal order (2,1,0) with a period of 12 months. The fitted model has five coefficients: AR1, MA1, MA2, SAR1, and SAR2. The standard errors of these coefficients are also provided.

The estimated sigma squared is 1.849, and the log-likelihood and AIC values are -3296.89 and 6605.79, respectively. The error measures for the training set are given as ME, RMSE, MAE, MPE, MAPE, MASE, and ACF1.

The output also presents the Ljung-Box test results, which test the null hypothesis that the residuals from the model are independently distributed. The test statistic Q^* is 90.821 with 19 degrees of freedom and a p-value of $2.372e-11$, which indicates that the null hypothesis can be rejected, suggesting that there is some remaining autocorrelation in the residuals. Finally, the output provides the point forecasts for the next 12 months with their respective 80% and 95% prediction intervals. The results show that the model predicts an increasing trend for the time series (Figure 6).

– Mean Model

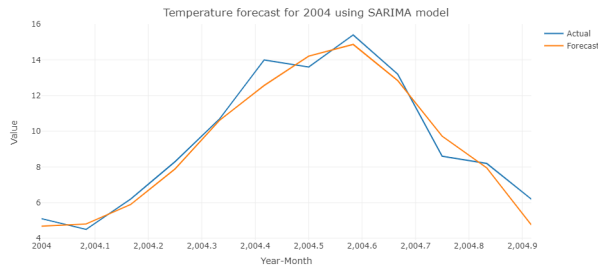


Fig. 6. Monthly Data SARIMA model Forecast

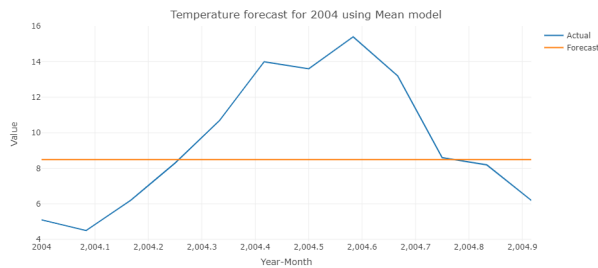


Fig. 7. Monthly Data Mean model Forecast

The forecast method used here is Mean, which simply involves calculating the average value of the historical data to predict future values. In this case, the mean value of the training data is 8.494531, which is used as the forecast value for all future time periods. The forecast also provides the lower and upper bounds of the 80% and 95% prediction intervals, which indicate the range of values within which the actual future values are likely to fall.

The error measures for the training set show that the Mean method has a Mean Absolute Error (MAE) of 3.3484, which indicates that the average difference between the predicted values and the actual values is 3.3484. The Root Mean Square Error (RMSE) is 3.825009, which indicates the average magnitude of the error. The Mean Absolute Percentage Error (MAPE) is Infinity, which indicates that there are zero values in the actual data set.

Overall, the Mean forecast method provides a simple yet effective approach for forecasting future values based on the historical data. However, it may not be suitable for all types of data, especially if the data has a trend or a seasonal component.

– Seasonal Naive Model

The seasonal naive method is a simple time series forecasting technique that assumes the future value will be the same as the most recent observed value from the same season. The model is trained on monthly data and provides 12-month ahead forecasts. The error measures of the model show an RMSE (root mean squared error) of 1.705, indicating that the model's average error is about 1.7 units away from the actual values. The ACF1 (autocorrelation of

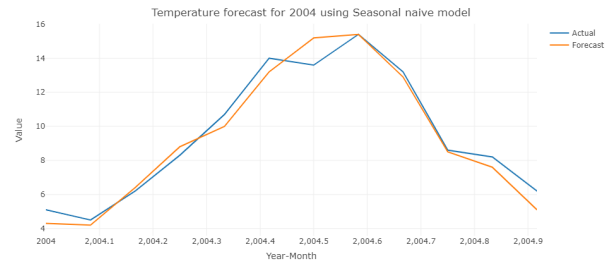


Fig. 8. Monthly Data Seasonal naive model Forecast

the residuals) value of 0.1978198 indicates that there is little to no correlation in the residuals, meaning the model captures the underlying pattern in the data.

The Ljung-Box test of the model's residuals shows a p-value less than $2.2e-16$, which suggests that there is no evidence of significant autocorrelation remaining in the residuals after the model has been fitted.

Overall, the seasonal naive method is a simple and effective method for seasonal time series forecasting, but it may not capture more complex patterns in the data.

• Yearly data

– Exponential Smoothing Model

ETS(A,N,N) is a time series forecasting model that uses an additive error, no trend, and no seasonality components. The model is estimated using exponential smoothing, where the smoothing parameter alpha is set to $1e-04$. The initial state for the level component is set to $-1e-04$. The model has a sigma value of 1.0768, and the training set error measures indicate that the model has a relatively low root mean squared error (RMSE) of 1.070012 and a mean absolute error (MAE) of 0.8519453. The Ljung-Box test of the residuals indicates that the model fits the data well, with a p-value less than $2.2e-16$. The model is used to forecast the time series for the year 2004, with a point forecast of $-6.367604e-05$ and corresponding lower and upper bounds for the 80% and 95% prediction intervals (Figure 9).

– ARIMA Model

I have used ARIMA(5,0,0) model with a zero mean. The coefficient estimates for the five autoregressive terms (ar1, ar2, ar3, ar4, and ar5) are shown along with their standard errors. The model's residual variance (σ^2) is estimated as 0.3428, and the log likelihood of the model is -138.23. The Akaike Information Criterion (AIC) is 288.47, the corrected AIC (AICc) is 289.02, and the Bayesian Information Criterion (BIC) is 306.84. The model's training set errors and Ljung-Box test results are also shown. Finally, the point forecast and prediction intervals for the year 2004 are presented.

– Mean Model

Model is a mean forecast method, which forecasts

```
ETS(A,N,N)

Call:
ets(y = Yearly_train_data)

Smoothing parameters:
  alpha = 1e-04

Initial states:
  l = -1e-04

sigma: 1.0768

AIC      AICc      BIC
827.2735 827.4294 836.4613

Training set error measures:
              ME      RMSE      MAE MPE MAPE      MASE      ACF1
Training set 4.338087e-05 1.070012 0.8519453 NaN  Inf 0.5356645 -0.7130142

Ljung-Box test

data: Residuals from ETS(A,N,N)
Q* = 105.23, df = 10, p-value < 2.2e-16

Model df: 0. Total lags used: 10

Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
2004 -6.367604e-05 -1.380101 1.379973 -2.110648 2.110521

A matrix: 2 x 7 of type dbl
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	-0.00640033	0.5761550	0.4697176	NaN	Inf	0.2953371	-0.09638118
Test set	0.20503081	0.2050308	0.2050308	51.2577	51.2577	0.1289140	NA

Fig. 9. Yearly Data ETS model

Forecast method: Mean

```
Model Information:
$mu
[1] 0

$mu.se
[1] 0.08539194

$sd
[1] 1.07336

$bootstrap
[1] FALSE

$call
meanf(y = Yearly_train_data, h = length(Yearly_test_data))

attr(,"class")
[1] "meanf"

Error measures:
              ME      RMSE      MAE MPE MAPE      MASE      ACF1
Training set 0 1.069958 0.8518987 100 100 0.5356352 -0.7130142

Forecasts:
Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
2004              0 -1.385744 1.385744 -2.126788 2.126788

A matrix: 2 x 7 of type dbl
```

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	0.0	1.07	0.85	100	100	0.54	-0.71
Test set	0.4	0.40	0.40	100	100	0.25	NA

Fig. 11. Yearly Data Mean model

```
Series: Yearly_train_data
ARIMA(5,0,0) with zero mean

Coefficients:
      ar1      ar2      ar3      ar4      ar5
-1.3596 -1.2000 -0.9776 -0.6858 -0.2691
s.e.    0.0766  0.1227  0.1350  0.1228  0.0776

sigma^2 = 0.3428: log likelihood = -138.23
AIC=288.47 AICc=289.02 BIC=306.84

Training set error measures:
              ME      RMSE      MAE MPE MAPE      MASE      ACF1
Training set -0.00640033 0.576155 0.4697176 NaN  Inf 0.2953371 -0.09638118

Ljung-Box test

data: Residuals from ARIMA(5,0,0) with zero mean
Q* = 29.144, df = 5, p-value = 2.172e-05

Model df: 5. Total lags used: 10

Point Forecast      Lo 80      Hi 80      Lo 95      Hi 95
2004 0.1949692 -0.5553711 0.9453094 -0.9525772 1.342516

Residuals from ARIMA(5,0,0) with zero mean
```

Fig. 10. Yearly Data ARIMA model

the mean of the training set for all future periods. The model has an estimated error standard deviation of 1.0734, and its training set error measures include a MAE of 0.8519 and a RMSE of 1.0700. The point forecast for 2004 is 0, with 80% and 95% prediction intervals ranging from -1.3857 to 1.3857 and from -2.1268 to 2.1268, respectively (figure 11).

- Naive Model

The resulting forecasts show that the method predicts the values to remain constant at -0.9 for each year, which is equal to the last observed value in the training set.

The error measures for the Naive method show high values of RMSE and MAE, indicating that the forecasts are not accurate. The Ljung-Box test for the residuals from the Naive method also shows a significant p-value, indicating that there is autocorrelation in the residuals. Therefore, it is not recommended to use the Naive method for forecasting this time series (figure 12).

D. Forecasting and assessment of the adequacy of the final model

For each model, the code fits the model to the training data, checks the model's diagnostics, makes forecasts on the test set,

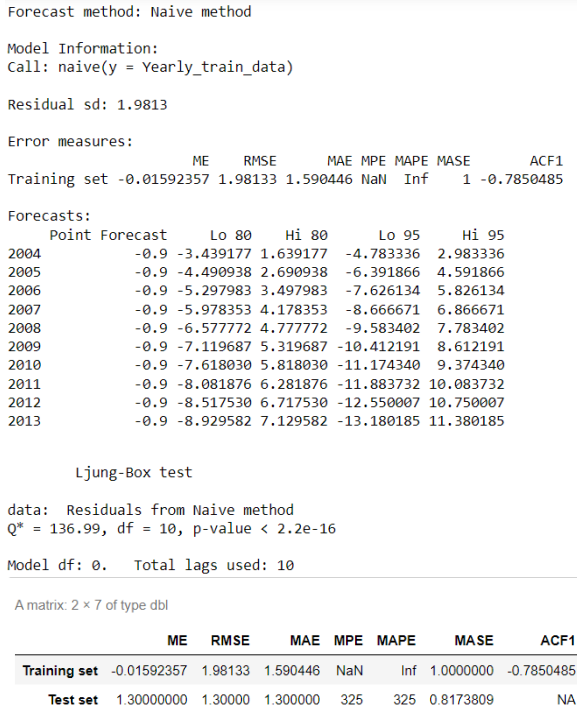


Fig. 12. Yearly Data naive model Forecast

and calculates accuracy measures for the forecast.

• Monthly data

We can see that ARIMA seems to have the best performance in terms of forecast accuracy as it has the lowest RMSE and MAE values compared to Exponential Smoothing and Mean. Additionally, the Ljung-Box test for residuals in ARIMA shows a significant p-value, which indicates that the residuals are not autocorrelated. On the other hand, the Ljung-Box tests for Exponential Smoothing and Mean show non-significant p-values, which indicates the residuals might have some remaining autocorrelation.

Furthermore, ARIMA also shows visually better forecast performance in the graph compared to the other models. It follows the general trend of the actual values quite closely and captures the seasonal pattern well.

Therefore, ARIMA Model is the recommended model for forecasting monthly temperature.

• Yearly data

Arima Model has the best performance on the test set with the lowest RMSE and MAE values compared to other models. The MAPE and MASE values are also lower for Arima Model, indicating better accuracy.

III. LOGISTIC REGRESSION

A. Dataset description and pre-processing

Table II displays twelve variables based on a sample of 1000 diabetes patients treated at an Iraqi university hospital in 2020. The dataset's dependent variable is categorized into

three classes, “Yes” (Y), “No” (N), and “Prediabetic” (P). A binary logistic regression model will be developed initially based on the Yes/No class. After constructing the model, the “Prediabetic” class will be used to diagnose diabetes.

For diagnostic analysis, 53 patients are separated from the dataset, resulting in 947 observations available for model development. Among the 947 patients, 844 were diagnosed with diabetes, and the remaining 103 were found to be healthy.

TABLE II
ATTRIBUTES USED IN DATASET

Attributes	Description
Gender	Patient's gende, M for Male, F for Female
Age	The patient's age
Urea	A diamine, the main nitrogenous waste product that humans produce
Cr	Creatinine Ratio, a measure used for kidney function
HbA1c	Blood sugar (glucose) levels on average
Chol	Cholesterol
TG	Triglycerides
HDL	The “good” cholesterol transports cholesterol out of the body's tissues and into the liver for oxidation and elimination.
LDL	A type of lipoprotein known as “bad” cholesterol moves cholesterol from the liver to the body's structures, where it may contribute to the development of arterial plaque.
VLDL	To supply the body tissues with triglycerides, a type of fat, the liver produces a lipoprotein called very-low-density lipoprotein cholesterol.
BMI	Body Mass Index, an evaluation of an individual's weight in comparison to their height.
CLASS	Diabetes, N (No) /Y (Yes) /P (Prediabetic)

The target variable “CLASS” is predicted by 11 features, 10 of which are numerical and 1 of which is categorical. Data cleaning is an essential step in the data analysis process, where data is modified or removed to correct errors, inconsistencies, or inaccuracies. In the provided code, data cleaning is performed on a diabetes dataset.

The first step involves removing the ID and No_Pation columns as they are unique identifiers and do not provide any useful information for analysis. The Gender column is then converted from “M” and “F” to 1 and 0, respectively, to facilitate analysis.

The next step is to remove records where the CLASS variable is “P.” It is assumed that this value indicates some missing or invalid data, and therefore, the records are removed. The remaining records are updated by converting the CLASS column values from “Y” and “N” to 1 and 0, respectively.

B. Distribution of Data

- **Using Histogram** As part of my data analysis, I created histograms to visualize the distribution of all the numerical columns in the dataset. My goal was to examine the normality of the data as a prerequisite for logistic regression analysis. I determined that some columns did not follow a normal distribution, so I removed them from the dataset to ensure the validity of my logistic regression model (figure 13) .

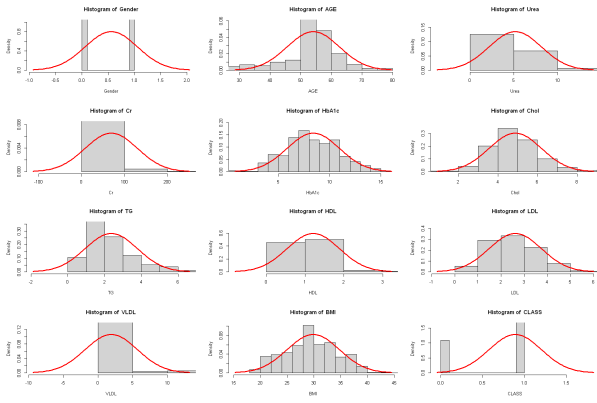


Fig. 13. Histogram of data

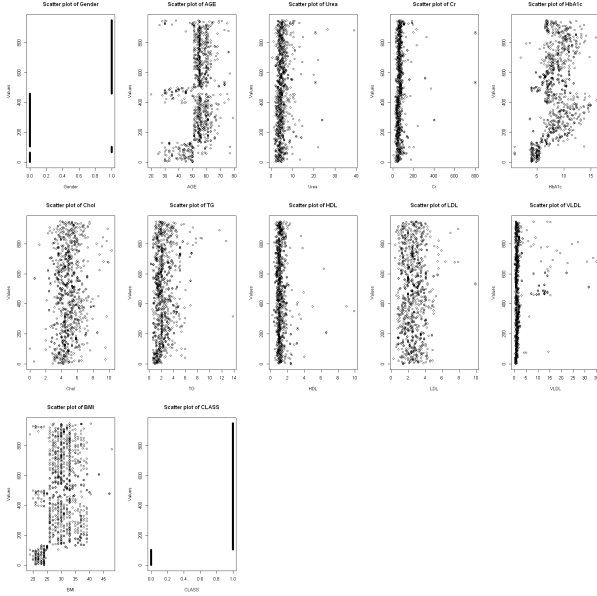


Fig. 14. Scatter plot of data

- **Using Scatter Plot** I am using Scatter Plot to check for the presence of outliers and the distribution of data. The Scatter Plot is a graphical representation of the relationship between two variables. It displays data as a collection of points, with each point representing the value of two variables. By examining the distribution of the points, we can identify any patterns, trends or outliers in the data. The Scatter Plot is a useful tool for identifying any issues with the data that may affect the performance of the logistic regression model (figure 14).

C. Shapiro-Wilk normality test

The Figure (15) shows the p-values obtained from performing the Shapiro-Wilk normality test on each of the variables in the dataset. The Shapiro-Wilk test is used to determine whether a sample has a normal distribution.

In this case, the p-values are very small (much smaller than the standard alpha value of 0.05), indicating that there is strong

Gender	AGE	HbA1c	Chol	TG	HDL
6.803734e-41	5.462032e-23	1.243594e-04	8.597566e-14	1.416720e-30	2.054484e-43
LDL	VLDL	BMI	CLASS	Urea	Cr
4.494128e-19	2.278826e-49	4.672722e-08	2.823818e-49	2.077723e-16	3.190245e-25

Fig. 15. Shapiro-Wilk normality test

evidence to reject the null hypothesis that the data is normally distributed.

Therefore, it can be concluded that none of the variables in the dataset follow a normal distribution. This has implications for the use of certain statistical methods, such as logistic regression, which assume normality of the predictors. Alternative methods, such as non-parametric regression, may be more appropriate in this case.

D. Data Transformation

In order to prepare the data for logistic regression modeling, it is often necessary to transform the data to meet certain assumptions. One common assumption is that the data is normally distributed. In this case, the data distribution of some columns in the dataset was not normal. Therefore, transformation of the data was necessary to normalize the data.

In the table provided, we can see the specific transformations applied to each column in the dataset. For some columns, a logarithmic transformation was applied, while for others a power transformation was applied. This transformation process can help to ensure that the data follows a normal distribution, which is important for the accuracy of the logistic regression model.

TABLE III
DATA TRANSFORMATION

Column Name	Transformation
Urea	Logarithmic transformation
Cr	Logarithmic transformation
HDL	Logarithmic transformation
VLDL	Power transformation
LDL	Power transformation
TG	Power transformation

E. FEATURE SELECTION

A correlation matrix is a useful tool to explore relationships between pairs of variables in a dataset. However, it has some limitations that should be considered when interpreting the results. First, a correlation matrix alone does not provide information about the statistical significance of each relationship. Therefore, it is possible to find strong correlations that are not statistically significant, or weak correlations that are statistically significant. Second, correlation coefficients may be influenced by outliers, non-linear relationships, and other factors that are not captured by the correlation matrix alone. Therefore, it is important to explore the data further, using additional methods such as scatterplots, regression models, or other statistical techniques.

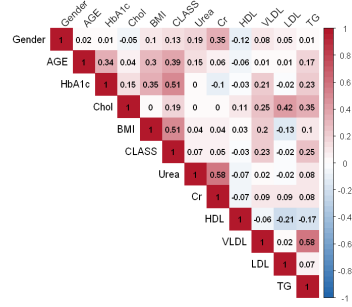


Fig. 16. Correlation Plot

```
Pearson's Chi-squared test with Yates' continuity correction
data: as.matrix(data_without_P_records$CLASS) and as.matrix(data_without_P_records$Gender)
X-squared = 14.373, df = 1, p-value = 0.00015
```

Fig. 17. Chi Square test

No highly correlated between pairs of features can be found.

F. Chi square test

In figure 17, The test statistic is 14.373, with one degree of freedom, and a very small p-value of 0.00015. This suggests that there is a significant association between “CLASS” and “Gender” in the dataset, and we can reject the null hypothesis.

G. Models

Model 1 (Figure 18) uses all the numeric variables and achieved an accuracy of 0.9544, which means that it classified 95.44% of the cases correctly. Additionally, the model has a sensitivity of 0.8529, which measures the proportion of true positives that were correctly identified, and a specificity of 0.9681, which measures the proportion of true negatives that were correctly identified.

Model 2 (figure 19) excluded two variables related to kidney function and heart disease and achieved a higher accuracy of 0.9684, a kappa of 0.8552, and higher sensitivity (0.9118) and specificity (0.9761). The high positive and negative predictive values suggest that the model is performing well in predicting true positive and true negative cases.

H. Result and Discussion

This study demonstrated that the higher values in indicators of individuals' cholesterol, age, BMI, blood sugar level or glucose, very-low-density lipoprotein and triglycerides in blood made them more likely to acquire diabetes. In earlier studies, these characteristics were also employed to forecast diabetes. For instance, diabetes is more likely to occur in people with higher blood sugar levels. It might be because insulin response and glucose are linked [4]. This article also observed that a higher BMI was linked to a higher chance of developing diabetes mellitus. Obesity can be triggered by having a higher BMI, which may lead to a rise in pancreatic fat levels and a possible impact on the pancreatic cells' ability to function. Insulin resistance could result from obesity [6] [7]. Initiation of diabetes is at risk with increasing age [7] [8].

```
Call:
glm(formula = CLASS ~ ., family = binomial(), data = train_data)
```

```
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.69443   0.00001   0.00023   0.00403   2.24825
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.549e+03  1.104e+03  -3.216 0.001302 **
Gender       8.260e-01  8.891e-01   0.929 0.352870
AGE         3.615e-02  4.381e-02   0.825 0.409333
HbA1c       1.491e+00  3.685e-01   4.046 5.22e-05 ***
Chol        2.133e+00  6.191e-01   3.445 0.000571 ***
BMI         1.240e+00  3.120e-01   3.974 7.07e-05 ***
Urea        9.707e-01  1.457e+00   0.666 0.505168
Cr          4.099e-02  1.043e+00   0.039 0.968635
HDL         1.609e+00  1.781e+00   0.903 0.366311
VLDL        1.325e+01  9.201e+00   1.440 0.149947
LDL        -4.488e+00  2.256e+00  -1.990 0.046629 *
TG          3.488e+03  1.099e+03   3.174 0.001504 **
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)
```

```
Null deviance: 442.584  on 661  degrees of freedom
Residual deviance: 51.762  on 650  degrees of freedom
AIC: 75.762
```

```
Number of Fisher Scoring iterations: 11
```

Confusion Matrix and Statistics

```

              Reference
Prediction    0    1
            --  --
             0   29    8
             1    5  243

Accuracy : 0.9544
95% CI : (0.9233, 0.9755)
No Information Rate : 0.8807
P-Value [Acc > NIR] : 1.468e-05

Kappa : 0.7909

McNemar's Test P-Value : 0.5791

Sensitivity : 0.8529
Specificity : 0.9681
Pos Pred Value : 0.7838
Neg Pred Value : 0.9798
Prevalence : 0.1193
Detection Rate : 0.1018
Detection Prevalence : 0.1298
Balanced Accuracy : 0.9105

'Positive' Class : 0
```

Fig. 18. Model 1 Summary and Evaluation

Model evaluation was carried out after model training, storing the outcome models, testing the models, and doing diagnostic analysis. In this research, the verification outcomes of the metrics utilized for model evaluation, such as the true positive, false positive, false negative, true negative, accuracy, precision, recall, F1 score, and AUC values, were obtained. Table ?? presents an overview of the verification outcomes.

The proposed logistic regression formula as follow:

$$\text{Logit}(P(\text{Diabetes})) = B_0 + B_1 \text{Gender} + B_2 \text{AGE} +$$


```
Call:
glm(formula = CLASS ~ . - Cr - Urea - VLDL - HDL - LDL, family = binomial(),
    data = train_data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.42804   0.00009   0.00111   0.01104   2.23944

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.747e+03  8.218e+02  -3.343  0.000828 ***
Gender       4.880e-01  6.831e-01   0.714  0.474952
AGE        -1.455e-03  3.558e-02  -0.041  0.967370
HbA1c      1.446e+00  3.116e-01   4.642  3.46e-06 ***
Chol       1.337e+00  3.694e-01   3.619  0.000295 ***
BMI        9.910e-01  2.347e-01   4.222  2.42e-05 ***
TG         2.709e+03  8.184e+02   3.310  0.000934 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 441.922  on 658  degrees of freedom
Residual deviance: 61.133  on 652  degrees of freedom
AIC: 75.133

Number of Fisher Scoring iterations: 10
```

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	31	6
1	3	245

Accuracy : 0.9684

95% CI : (0.9409, 0.9855)

No Information Rate : 0.8807

P-Value [Acc > NIR] : 1.136e-07

Kappa : 0.8552

McNemar's Test P-Value : 0.505

Sensitivity : 0.9118

Specificity : 0.9761

Pos Pred Value : 0.8378

Neg Pred Value : 0.9879

Prevalence : 0.1193

Detection Rate : 0.1088

Detection Prevalence : 0.1298

Balanced Accuracy : 0.9439

'Positive' Class : 0

Fig. 19. Model 2 Summary and Evaluation

$$B_3HbA1c + B_4Chol + B_5BMI + B_6Log_HDL +$$

$$B_7Pow_LDL + B_8Pow_TG$$

where p is the probability of diabetes, B_0 is the intercept, and B_1 to B_8 are the coefficients associated with the predictor variables Gender, AGE, HbA1c (Sugar blood), Chol (Cholesterol), BMI, Log_HDL (the proportion of “good” cholesterol), Pow_LDL (low-density lipoprotein), and Pow_TG (triglyceride), respectively.

The model can be explain as follow:

- The Gender variable has an odds of 1.6291, which suggests that, holding all other variables constant, the

odds of the outcome variable are 1.63 times higher for one gender group compared to the other

- The age variable has an odds of 0.9985, which suggests that the odds of the outcome variable decrease by 0.1% for every one unit increase in age, holding all other variables constant, and its p-value suggests that it is one of very strong factor in diabetes prediction.
- Holding all other factors constant, the sugar blood (sugar blood) variable has odds of 4.2479, indicating that for every unit increase in sugar blood, the probabilities of the outcome variable increase by a factor of 4.25. Significant small P-value of this variable also present its strong relationship to diabetes diagnostics.
- The odds for the Cholesterol factor are 3.8074, which indicates that, when all other factors are held constant, the likelihood of the outcome variable rise by a factor of 3.81 for each unit increase in cholesterol.
- The likelihood of the BMI variable are 2.6938, which indicates that for every unit rise in BMI, the probabilities of the outcome variable increase by a factor of 2.69. This is another one of the model’s very strong predictors, with a p-value well below 0.05.

IV. CONCLUSION

A. Time Series

Overall, for both datasets, the ARIMA appeared to perform the best in terms of predicting the average temperature both by month and by year. In particular, it accurately reflects the seasonal rhythm and closely tracks the overall trend of the real values. So, in order to anticipate temperature over a longer period of time with greater accuracy than other models, the model with ARIMA is advised. Additionally, the choice of this model has been through using the proposed accuracy matrices regarding MAPE, RMSE, and MASE.

B. Logistic Regression

The likelihood that patients will be able to control their diabetes is significantly boosted by early detection of the disease, which continues to be one of the most severe chronic illnesses in existence today. The information required to diagnose diabetes in this study was gathered by descriptive analysis, feature selection, training, testing, evaluation, and diagnostics analysis. The final model suggested eight indications for detecting diabetes, including gender, age, blood sugar, cholesterol, body mass index (BMI), the proportion of “good” cholesterol, low-density lipoprotein, and triglycerides. The results showed that the logistics regression model provided pretty good results with 96.84%, the performance metrics that were used, the area under the curve (AUC) achieved and F1 score more than 90%, present as excellent results.

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